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27 DECEMBER 1961
SUPERSEDING
MIL-F-11150(CmlC)
10 MAY 1951

MILITARY SPECIFICATION

FILTER, PARTICULATE, 150 CFM, ABC-M9

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers a particulate filter designed to remove particulate matter from atmospheric air.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–282 — Filter Units, Protective Clothing, Gas Mask Components and Related Products; Performance Test Methods.

DRAWINGS

CHEMICAL CORPS

C5–19–922 — Filter, Particulate, 150 CFM, ABC–M9 — Bill of Material.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies of these freight classification rules should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 *Materials.* All materials cited on Drawing C5–19–922 shall conform to the spec-

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ifications listed thereon and to the specific characteristics set forth in the drawings.

3.1.2 *Components.* All components of the item conform to the specifications and drawings listed on drawing C5–19–922.

3.2 **Manufacture.** The filter shall be manufactured to conform to drawing D5–19–706 and to the drawings listed on drawing C5–19–922.

3.3 **Resistance to airflow.** The resistance to airflow through the filter shall not exceed 1.0 inch of water at an airflow of 150 cubic feet per minute (cmf) when tested in accordance with 4.6.1.

3.4 **DOP smoke penetration.** The DOP smoke penetration of the filter shall not exceed 0.03 percent when tested as specified in 4.6.1.

3.5 **Resistance to rough handling.** The filter shall comply with the resistance to airflow and DOP smoke penetration requirements of 3.3 and 3.4 after rough handling as specified in 4.6.2.

3.6 **Workmanship.** Damage, such as bent, warped, or burred metal sections, and contamination of the surface with foreign substances, such as oily or viscous liquids, shall be indicative of bad workmanship and not allowed.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of of this specification shall govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the particulate filters produced by one manufacturer at one plant from the same material under essentially the same manufacturing conditions and offerered for acceptance at one time. Not more than one lot of filter material shall be represented in any one lot of particulate filters.

4.4 **Sampling.**

4.4.1 *For examination.* Sampling shall be conducted in accordance with Standard MIL–STD–105.

4.4.2 *For test.* From each lot, 20 filters shall be selected at random for testing as specified in 4.6.2.

4.5 **Examination.** Sample filters shall be examined in accordance with the classification of defects and with Standard MIL–STD–105.

4.5.1 *Classification of defects.*

4.5.1.1 *Filter, particulate, 150 cfm, ABC–M9 assembly (dwg D5–19–706, which is listed on dwg C5–19–922).*

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Categories	Defects	Acceptance standards
Critical:*		
1	Resistance to airflow.	4.6.1
2	DOP smoke penetration.	4.6.1
Major:	AQL 1.0 percent defective	
101	Surface under gasket uneven, bent, or poorly finished.	
102	Gasket improperly cemented in place or damaged.	
103	Marking incorrect, missing, or illegible.	
Minor:	AQL 1.5 percent defective	
201	Workmanship (see 3.6).	

* Each filter in the lot shall be inspected for these defects.

4.6 Tests. Tests shall be conducted as follows:

4.6.1 *Resistance to airflow and DOP smoke penetration.* The filter shall be tested for DOP smoke penetration and resistance to airflow in accordance with method 102.9.1 of Standard MIL–STD–282, using an airflow of 150 cfm.

4.6.2 *Resistance to roudgh handling.* The filters with pleats in a vertical position shall be rough-handled for 15 minutes in accordance with method 105.9 of Standard MIL–STD–282 and then tested for DOP smoke penetration and resistance to airflow in accordance with 4.6.1.

4.7 **Acceptance/rejection criteria.** If any filter fails when tested in accordance with 4.6.2, the lot shall be rejected.

5. PREPARATION FOR DELIVERY (see 6.5)

5.1 **Packaging, level C.** Packaging shall be in accordance with the manufacturer's commercial practice.

5.2 **Packing, level C.** Filters, packaged as specified in 5.1, shall be packed in a manner to insure safe arrival at first destination for immediate use. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.** In addition to any marking required by the contract or order, shipments shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Intended use.** The particulate filter covered by this specification is intended to be used in the ABC–M6 filter unit.

6.2 **Ordering data.** Procurement documents should specify the title, number, and date of this specification.

6.3 **International interest.** Certain provisions of this specification are the subject of international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

6.4 **Manufacturing suggestions.** Filters manufactured along the following lines have been satisfactory:

Pleat enough filter material (MIL–F–13785) on an appropriate pleating machine to give the desired compactness when fitted into the filter frame. It is suggested that fiberboard separators be temporarily placed between pleats and that the pleated filter material be placed in a jig. The jig should

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hold the pleated filter material in the same state of compactness as will be required in the filter frame and at the same time allow access to both ends and one side of the pleated element. The temporary fiberboard separators should be withdrawn and replaced by comb type separators. It is desirable that the comb type separators be formed on the day they are incorporated into the filter element. To replace separators on the opposite side, the filter element will have to be turned over in the jig. When all comb type separators are in place, the ends should be trimmed to the desired length with a high speed band saw. Determination of the desired number of pleats at the start of the operation can be determined by trial and error methods; however, once the number is determined it should be strictly adhered to as deviation from the correct force compressing the filter element will result in detrimental complications.

Coat inside of filter frame with sealing adhesive and insert pleated element into frame. On both sides of filter, seal entire seam between filter element and frame with sealing adhesive.

6.5 **Packaging and packing levels.** When other than level C packaging and packing is desired, the repair parts packaging data sheet covering the ABC–M9, 150 cfm particulate filter should be used. (Copies of repair parts packaging data sheets may be obtained from the Commanding Officer, ATTN: Chief, Requirements and Specification Office, U. S. Army Chemical Arsenal, Army Chemical Center, Maryland.)

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—CmlC
Navy—Docks
International interest (see 6.3)

Review interest:
Navy—Docks

User interest:
Navy—MC

Preparing activity:
Army—CmlC

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